

# Math 340: Elementary Matrix and Linear Algebra

## FINAL EXAM

Saturday, December 16th, 2023, 12:25pm-2:25pm

Please circle your Instructor and your Discussion Session (date and start time given) in the table below.

| Dr. Kaitlyn Phillipson | Dr. Shengwen Gan   | Dr. Joshua Mundinger | Dr. Feng Zhu    |
|------------------------|--------------------|----------------------|-----------------|
| Joey (Yu), T 3:30 PM   | Zaidan, M 12:05 PM | Katerina, M 7:45 AM  | Anna, T 1:20 PM |
| Joey (Yu), T 4:35 PM   | Zaidan, W 12:05 PM | Katerina, M 8:50 AM  | Anna, T 2:25 PM |
| Joey (Yu), R 3:30 PM   | Daniel, M 12:05 PM | Ruitao, W 7:45 AM    | Anna, R 2:25 PM |
| Yaoxian, R 4:35 PM     | Daniel, W 1:20 PM  | Ruitao, W 8:50 AM    |                 |
| Hongyu, R 9:55 AM      | Dewei, M 3:30 PM   | Jiaqi, M 11:00 AM    |                 |
| Hongyu, T 9:55 AM      | Dewei, W 3:30 PM   | Jiaqi, W 12:05 PM    |                 |
| Hongyu, R 8:50 AM      | Dewei, M 2:25 PM   | Ellie, W 1:20 PM     |                 |
|                        | Joel, W 4:35 PM    | Ellie, M 1:20 PM     |                 |

Name: Nicolas Bourbaki Wisc email: \_\_\_\_\_

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: \_\_\_\_\_

### READ THE FOLLOWING INFORMATION.

- This is a 120-minute exam. It consists of 11 problems for a total of 100 points; the exam is 8 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- Notation reminders:  $M_{mn}$  is the vector space of  $m \times n$  real matrices with standard matrix addition and scalar multiplication,  $P_d$  is the vector space of polynomials (in  $x$ ) of degree at most  $d$  with standard polynomial addition and scalar multiplication,  $\mathbf{0}$  denotes the zero vector in a vector space  $V$ .  $\mathbb{R}^n$  uses the standard vector addition/scalar multiplication unless otherwise noted.

**DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.**

1. (12 points, 2 points each) Are each of the following true or false statements? Fill in the bubble next to the correct answer. Justification is not necessary.

a.) Suppose  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  is a linear transformation defined by  $T(\mathbf{x}) = A\mathbf{x}$ , for  $A$  a  $3 \times 3$  matrix with  $\det(A) = -1$ . Then  $T$  is an isomorphism.

- ☒ True  
☐ False

b.) If  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$  is an orthonormal set, then  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$  is linearly independent.

- ☒ True  
☐ False

c.) For the inner product on  $\mathbb{R}^2$  defined by  $\left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle = u_1 v_1 + 2u_2 v_2$ , the vectors  $\mathbf{u} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

and  $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$  are orthogonal.

- ☐ True  
☒ False

$$\langle \vec{u}, \vec{v} \rangle = (3)(1) + 2(-1)(3) = -3 \neq 0.$$

d.) If  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$  is a linearly independent set, then so is  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3\}$ .

- ☒ True  
☐ False

e.) If  $B = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$  and  $D = (\mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_1)$  are two ordered bases for a vector space  $V$ , then the

change matrix  $P_{D \leftarrow B} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ .

- ☐ True  
☒ False

$$C_D(\vec{b}_1) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ X.}$$

f.) It is possible to build an onto (surjective) linear transformation from  $\mathbb{R}^3$  to  $P_3$ .

- ☐ True  
☒ False

2. (12 points total)

a. (4 points)  $A$  and  $B$  are similar matrices. Find  $k$ . Show all work.

similar

$$A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -2 \\ 0 & -1 & 1 \end{bmatrix}, B = \begin{bmatrix} 4 & 0 & 1 \\ 0 & -2 & k \\ -1 & 2 & 3 \end{bmatrix}$$

$$\det(A) = \det(B) = 4(-6 - 2k) + 1(0 - 2) = -24 - 8k - 2 = -26 - 8k$$

$$\begin{aligned} &1(-4 - 1) + 1(4 - 1) \\ &\quad \parallel \\ &-5 + 3 = -2 \end{aligned}$$

$$-2 = -26 - 8k$$

$$24 = -8k \Rightarrow$$

$$\boxed{k = \frac{24}{-8} = -3}$$

b. (8 points) Suppose  $C, D, E, F$  are  $3 \times 3$  matrices, where  $\det(C) = 3$  and  $\det(D) = -2$ .(i) (4 points) If  $C^{-1}D^T E = I_3$ , find  $\det(E)$ . Show all work.

$$\det(C^{-1}D^T E) = \det(I_3) = 1$$

$$\parallel$$

$$\frac{1}{\det(C)} \det(D) \det(E) = 1$$

$$-\frac{2}{3} \det(E) = 1 \Rightarrow \det(E) = -\frac{3}{2}$$

(ii) (4 points) Suppose  $F$  is obtained from  $C$  via the row operation  $r_1 + 4r_2 \rightarrow r_1$  (Hint: read this row operation VERY carefully). What is  $\det(F)$ ? Briefly explain your answer.

This mult. the det. by 4.

$$\det(F) = (3)(4) = 12.$$

3. (7 points) Suppose  $L : P_2 \rightarrow P_1$  is a linear transformation with  $M_{DB}(L) = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix}$  for ordered bases  $B = (1, x^2, x)$  and  $D = (x+2, x-1)$ . Find  $L(2x^2 - 3x + 1)$ .  
(note: your final answer should be a polynomial, not a column vector). Show all work.

$${}_B(2x^2 - 3x + 1) = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 2+2-9 \\ -2-6 \end{bmatrix} = \begin{bmatrix} -5 \\ -8 \end{bmatrix}$$

↓ D

$$-5(x+2) - 8(x-1) = \boxed{-13x - 2}$$



4. (10 points) Consider  $L : P_1 \rightarrow P_1$  by  $L(ax + b) = bx + 2a - b$ . You can assume that  $L$  is a linear transformation.

a. (4 points) Find  $M_B(L)$  using basis  $B = (1, x)$ . Show all work.

$$L(1) = x - 1 \rightarrow \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$L(x) = 2 \rightarrow \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$M_B(L) = \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}$$

b. (3 points) Is  $L$  an isomorphism? (hint: use your answer to part (a)). Justify your answer.

$$\det(M_B(L)) = -1 \cdot 0 - 2 = -2 \neq 0.$$

Yes! The matrix is invertible, so

$L$  is an isomorphism

c. (3 points) Is  $L$  diagonalizable? (hint: use your answer to part (a)). Justify your answer.

$$\det(\lambda I - M_B(L)) = \det \begin{bmatrix} \lambda + 1 & -2 \\ -1 & \lambda \end{bmatrix} = (\lambda + 1)(\lambda) - 2$$

Yes! The matrix is

diagonalizable, so  $L$  is diagonalizable.

$$= \lambda^2 + \lambda - 2 = 0$$

$$(\lambda - 1)(\lambda + 2) = 0$$

$$\lambda = -2, 1$$

2 distinct eigenvalues!

5. (7 points) Consider  $M_{22}$  and the subset  $W$  of matrices  $A$  where  $A^2 = A$ , i.e.

$$W = \{A \text{ in } M_{22} \mid A^2 = A\}$$

Is  $W$  a subspace of  $M_{22}$ ? Fully justify your answer.

no, consider:  $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$   $A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A$ . is in  $W$ .

But  $2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}^2 = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \neq \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ , so this is not closed under scalar add.

Also not closed under add:

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$  (same reason)

6. (7 points total) Consider the subspace  $W$  of  $\mathbb{R}^4$  spanned by the set

$$\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 5 \\ 0 \end{bmatrix} \right\}$$

a. (3 points) What is the dimension of  $W$ ? Show your work, and briefly justify your answer.

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 3 & 2 & 5 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 3 & 2 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{-3r_2 + r_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}r_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3 pivot =  $\dim(W) = 3$ ,

b. (2 points) Is  $W$  isomorphic to  $M_{22}$ ? Justify your answer in one sentence.

$$\text{no, } \dim(M_{22}) = 4, \text{ and } 4 \neq 3.$$

c. (2 points) For which  $d$  is  $W$  isomorphic to  $P_d$ ? Justify your answer in one sentence.

$$\dim(P_d) = d + 1 = 3 \quad d = 2.$$

$$P_2 \text{ b/c } \dim(P_2) = 3 = \dim(W).$$

7. (8 points) Suppose  $A = \begin{bmatrix} -2 & 4 \\ 3 & 2 \end{bmatrix}$  and  $C^{-1} = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix}$ . If  $B$  is a  $2 \times 2$  matrix with

$C(A+B) = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$ , find  $B$ . (Hint: You do NOT need to find  $C$  to solve this problem). Show all work.

$$B = \overset{C^{-1}}{\begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix}} \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix} - \overset{A}{\begin{bmatrix} -2 & 4 \\ 3 & 2 \end{bmatrix}}$$

$$= \begin{bmatrix} -3 & 15 \\ 0 & 6 \end{bmatrix} - \begin{bmatrix} -2 & 4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 11 \\ -3 & 4 \end{bmatrix}$$

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4x4 system

$$C = \begin{bmatrix} -\frac{4}{3} & \frac{5}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix} \leadsto \begin{bmatrix} -\frac{4}{3} & \frac{5}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix} \left( \begin{bmatrix} -2 & 4 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} \frac{19}{3} - \frac{2a}{3} + \frac{5c}{3} & \frac{2}{3} - \frac{2b}{3} + \frac{5d}{3} \\ -\frac{5}{3} + \frac{a}{3} - \frac{c}{3} & \frac{2}{3} + \frac{b}{3} - \frac{d}{3} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$$

↓  
solve system...

$$a = -1 \quad b = 11$$

$$c = -3 \quad d = 4$$

8. (10 points total) (This problem has overflow room on next page if needed)

For this problem, use  $A$  and the reduced row echelon form of  $A$ ,  $RREF(A)$ , to answer the following questions.

$$A = \begin{bmatrix} 1 & 3 & 1 & 6 & -1 \\ -1 & 0 & 5 & 7 & 2 \\ 0 & 2 & 4 & 8 & 2 \\ 2 & 4 & -2 & 5 & -6 \\ 1 & 2 & -1 & 1 & 0 \end{bmatrix}, \quad RREF(A) = \begin{bmatrix} 1 & 0 & -5 & 0 & -16 \\ 0 & 1 & 2 & 0 & 9 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 10 \\ a \\ a \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 - 10 \\ a + 4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Answer each question with a brief explanation and/or work. An answer with no work/explanation will NOT receive full credit.

a.) (2 points) Is  $A$  invertible?

No, b/c  $RREF(A)$  is not  $I_5$ .  
 or  $\text{rank}(A) = 3 \neq 5$  or  $\det(A) = \det(RREF) = 0$   
 or...  
 b/c row ops. don't make a det. zero.

b.) (2 points) For what value of  $a$  (if any) is  $\mathbf{x} = \begin{bmatrix} 10 \\ a \\ 2 \\ 0 \\ 0 \end{bmatrix}$  a solution to  $A\mathbf{x} = \mathbf{0}$ ?

(see above)  
 $a + 4 = 0 \Rightarrow a = -4$

c.) (2 points) Is  $\lambda = 0$  an eigenvalue of  $A$ ?

Yes! b/c  $\vec{x} = \begin{bmatrix} 10 \\ 4 \\ 2 \\ 0 \\ 0 \end{bmatrix}$  is an eigenvector of  $A$  associated to  $\lambda = 0$ .

d.) (2 points) Suppose  $T : \mathbb{R}^5 \rightarrow \mathbb{R}^5$  is the linear transformation  $T(\mathbf{x}) = A\mathbf{x}$  (yes, this is the same matrix  $A$  as above). Give the dimension of the kernel of  $T$ .

$\dim \ker T = \# \text{ of free variables in } RREF(A)$   
 $= 2$

e.) (2 points) Suppose  $T : \mathbb{R}^5 \rightarrow \mathbb{R}^5$  is the linear transformation  $T(\mathbf{x}) = A\mathbf{x}$  (yes, this is the same matrix  $A$  as above and in part (d)). Give a basis for the image of  $T$ .

pick up the pivots.

$$\begin{bmatrix} 1 \\ -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \\ 8 \\ 5 \\ 1 \end{bmatrix}$$

(Overflow room for Problem 8, if needed):

9. (8 points) Consider  $\mathbb{R}^3$  with the standard inner product (also known as the dot product), and consider the set  $W$  of vectors which are orthogonal to  $\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$ . Find a basis for  $W$ . (You may assume that  $W$  is a subspace of  $\mathbb{R}^3$ .)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = x - 2z = 0$$

$x = 2z$ ,  $y$  can be anything.

$$\begin{bmatrix} 2z \\ y \\ z \end{bmatrix} = z \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Basis:  $\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

10. (7 points total) Consider the following set of polynomials in  $P_2$ :

$$S = \{x^2 - 2x + 4, x - 2x^2 + 1, 12 - 6x^2\}$$

a. (4 points) Does  $p = x^2 - 2x - 1$  lie in the span of  $S$ ? Show all work.

$ax^2 + bx + c \rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix}$  isom.

Solve

$$\left[ \begin{array}{ccc|c} 1 & -2 & -6 & 1 \\ -2 & 1 & 0 & -2 \\ 4 & 1 & 12 & -1 \end{array} \right] \xrightarrow{\substack{2r_1 + r_2 \rightarrow r_2 \\ -4r_1 + r_3 \rightarrow r_3}} \left[ \begin{array}{ccc|c} 1 & -2 & -6 & 1 \\ 0 & -3 & -12 & 0 \\ 0 & 9 & 36 & -5 \end{array} \right]$$

$\swarrow -\frac{1}{3}r_2 \rightarrow r_2$

$$\left[ \begin{array}{ccc|c} 1 & -2 & -6 & 1 \\ 0 & 1 & 4 & 0 \\ 0 & 9 & 36 & -5 \end{array} \right] \xrightarrow{9r_2 + r_3 \rightarrow r_3} \left[ \begin{array}{ccc|c} 1 & -2 & -6 & 1 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & -5 \end{array} \right]$$

inconsistent!

No,  $p$  does not lie in the span.

b. (3 points) Does  $S$  form a basis for  $P_2$ ? Completely justify your answer.

No, it doesn't span all of  $P_2$ .  $x^2 - 2x - 1$  doesn't lie in the span.



11. (12 points total) (This problem has overflow room on next page if needed)

Consider the vector space  $P_1$ , and define the function

$$\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x)dx$$

for  $p(x), q(x)$  in  $P_1$ .

You can assume that  $\langle p(x), q(x) \rangle$  is an inner product for  $P_1$ .

a. (3 points) Compute  $\langle x+2, x \rangle$ . Show all work.

$$\begin{aligned} \langle x+2, x \rangle &= \int_0^1 (x+2)x dx = \int_0^1 x^2 + 2x dx = \left. \frac{x^3}{3} + x^2 \right|_0^1 = \boxed{\frac{1}{3} + 1} \\ &= \boxed{\frac{4}{3}} \end{aligned}$$

b. (2 points) Using this inner product  $\langle p(x), q(x) \rangle$ , is  $x+1$  a unit vector?

$$\begin{aligned} \langle x+1, x+1 \rangle &= \int_0^1 (x+1)^2 dx = \int_0^1 x^2 + 2x + 1 dx = \left. \frac{x^3}{3} + x^2 + x \right|_0^1 \\ &= \frac{1}{3} + 1 + 1 \neq 1. \end{aligned}$$

No.

c. (3 points) Using this inner product  $\langle p(x), q(x) \rangle$ , find the distance between  $x-2$  and  $2x+3$ . You do not need to simplify your final answer. Show all work.

$$\begin{aligned} 2x+3 - (x-2) &= x+5. \\ \|x+5\| &= \sqrt{\int_0^1 (x+5)^2 dx} = \sqrt{\int_0^1 x^2 + 10x + 25 dx} = \sqrt{\left. \frac{x^3}{3} + 5x^2 + 25x \right|_0^1} \\ &= \sqrt{\frac{1}{3} + 5 + 25} \end{aligned}$$

d. (4 points) Still using this inner product  $\langle p(x), q(x) \rangle$ , determine for which value(s) of  $a$  the vector  $ax-4$  is orthogonal to  $3x+2$ .

$$\begin{aligned} 0 &= \langle ax-4, 3x+2 \rangle = \int_0^1 (ax-4)(3x+2) dx = \int_0^1 3ax^2 + 2ax - 12x - 8 dx \\ &= \left. \frac{3ax^3}{3} + \frac{2ax^2}{2} - \frac{12x^2}{2} - 8x \right|_0^1 = a + a - 6 - 8 = 0 \\ 2a - 14 &= 0 \leftarrow \\ \Rightarrow a &= 7. \end{aligned}$$

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(Overflow room for Problem 11, if needed)

Scratch paper page! (DO NOT REMOVE-MUST BE TURNED IN)

Back of scratch paper page! (DO NOT REMOVE-MUST BE TURNED IN)